Qn	Solution
6(a)(i)	The mean KE of a gas molecule is directly proportional to the thermodynamic temperature of the gas. Since the gases in the two containers have the same temperature, the ratio: $\dfrac{\text{mean kinetic energy of a gas molecule in container A}}{\text{mean kinetic energy of a gas molecule in container B}} = 1$
6(a)(ii)	$PV = NKT$ $N = \dfrac{PV}{KT}$ $\dfrac{\text{number of gas molecules in container A}}{\text{number of gas molecules in container B}}$ $= \dfrac{N_A}{N_B}$ $= \dfrac{\left(\dfrac{P_A V_A}{KT_A}\right)}{\left(\dfrac{P_B V_B}{KT_B}\right)}$ $= \dfrac{V_A}{V_B}$ $= \dfrac{V}{4V}$ $= 0.25$
6(a)(iii)	Since $\dfrac{1}{2}m\langle c^2\rangle = \dfrac{3}{2}kT$ $\sqrt{\langle c^2\rangle} = \sqrt{\dfrac{3kT}{m}}$ $\dfrac{\sqrt{\langle c_A{}^2\rangle}}{\sqrt{\langle c_B{}^2\rangle}} = \dfrac{\sqrt{\dfrac{3kT_A}{m_A}}}{\sqrt{\dfrac{3kT_B}{m_B}}}$ $= \sqrt{\dfrac{m_B}{m_A}}$ $= \sqrt{\dfrac{2m}{m}}$ $= 1.4$

Qn	Solution
6(b)	$\dfrac{\sqrt{\langle c_A{}^2\rangle}}{\sqrt{\langle c_B{}^2\rangle}} = \sqrt{2}$ $\sqrt{\langle c_A{}^2\rangle} = \sqrt{2}\left(\sqrt{\langle c_B{}^2\rangle}\right)$ $= \sqrt{2}(940)$ $= 1329.36 \text{ m s}^{-1}$ $\dfrac{1}{2}m\langle c^2\rangle = \dfrac{3}{2}nRT$ When $n = 1$ $\dfrac{1}{2}M\langle c^2\rangle = \dfrac{3}{2}RT$ For gas A, $\dfrac{1}{2}M_A\langle c_A{}^2\rangle = \dfrac{3}{2}RT_A$ $\dfrac{1}{2}\left(4.0\times10^{-3}\right)(1329.36)^2 = \dfrac{3}{2}(8.31)T_A$ $T_A = 280 \text{ K}$
7(c)	The electric field strength at a point in an electric field is defined as the electric